Enrollment No._____

GANPAT UNIVERSITY

B.Sc.IT(CS) Sem-III Regular Nov-Dec 2024 (Remedial)

U63A1LOS : LINUX SERVER OPERATING SYSTEM

Time: 3 Hours | Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BTL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy**
1	Differentiate between Linux and Unix.	06	CO1	BTL4
2	Explain Regular Expression in detail.	06	CO2	BTL6
3	Explain FTP in brief.	06	CO4	BTL6
4	Describe the **rm** and **cp** commands along with their options for deleting and copying files.	06	CO3	BTL5
5	Explain the Components of the Linux operating system with figure.	06	CO1	BTL6
6	Write a short note on shadow file.	06	CO5	BTL3
7	Analyze the Input Redirection, Output Redirection and Error Redirection.	06	CO4	BTL5
Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy**
1	Explain all working with directories commands in brief.	06	CO2	BTL6
2	Write a short note on File Permission.	06	CO1	BTL3
3	Explain the following filters : **grep, wc, sed, uniq, cut**	06	CO2	BTL5
Q.3	**Answer the following: (Any Two)**	**12**		
1	Differentiate FTP vs SFTP.	06	CO4	BTL4
2	Write a short note on Absolute and Relative paths.	06	CO3	BTL3
3	What is Shell ? Explain different types of Shell.	06	CO3	BTL5
Q.4	**Answer the following: (Any One)**	**06**		
1	Write a short note on User Management.	06	CO5	BTL3
2	Discuss Encryption with crypt in detail.	06	CO5	BTL5

**** End of the Paper ****

Enrollment No._____

GANPAT UNIVERSITY

B. Sc.IT (Cyber Security) Sem-III Regular/Remedial Nov-Dec 2024

U63A2DMS DATABASE MANAGEMENT SYSTEM

Time: 3 Hours | Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BTL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy**
1	Elaborate Three Levels of the Architecture.	06	CO1	BTL4
2	Illustrate on Dr. E. F. Codd rules in details.	06	CO1	BTL4
3	Apply Functional Dependency with example.	06	CO2	BTL3
4	Classify the TCL SQL commands in details.	06	CO3	BTL4
5	Elaborate the datatype of SQL.	06	CO3	BTL5
6	Evaluate Subquery in SQL in details.	06	CO4	BTL5
7	Evaluate the spooling concept in details.	06	CO5	BTL5
Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy**
1	Analyze role of database administration.	06	CO1	BTL4
2	What is E-R diagram? Explain relationship of E-R-diagram in details.			BTL2/BTL4
3	What is normalization? Analyze first and second normal form in details	06	CO2	BTL2/BTL4
Q.3	**Answer the following: (Any Two)**	**12**		
1	What are constraints? Analyze the UNIQUE key and NOTNULL key in brief.	06	CO3	BTL2/BTL4
2	Classify the Aggregate Functions with example.	06	CO4	BTL4
3	Analyze the set operator in details.	06	CO4	BTL4
Q.4	**Answer the following: (Any One)**	**06**		
1	Demonstrate index and sequence in details.	06	CO5	BTL3
2	Design normalizes database design with ER relationship of Library management system.			BTL6

**** End of the Paper ****

Enrollment No._____

GANPAT UNIVERSITY

B.Sc.IT(CS) Sem-III Regular Nov-Dec 2024 (Remedial)

U63A3OCP OBJECT ORIENTED CONCEPTS AND PROGRAMMING

Time: 3 Hours | Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BTL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2-Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I

Q-1	Answer the following. (Any 5)	Marks 30	CO	Bloom's Taxonomy Pattern
1	Explain the different types of inheritance with suitable example.	06	CO2	BTL5
2	Explain multithreading in Java including thread life cycle and thread creation methods.	06	CO2	BTL4
3	How do you implement generic? Explain with example.	06	CO3	BTL5
4	Explain the Hashtable class in Java.	06	CO3	BTL1
5	Write a simple Java program using ServerSocket to listen for and accept connections from clients.	06	CO4	BTL6
6	Describe the process of passing Lambda Expressions as arguments to methods in Java with example.	06	CO4	BTL1
7	Discuss different uses of final keyword with suitable example.	06	CO1	BTL3

Section – II

Q-2	Answer the following (Any Two)	Marks 12	CO	Bloom's Taxonomy Pattern
1	Discuss the features Java programming language.	06	CO1	BTL2
2	State the differences between String and StringBuffer class.	06	CO1	BTL3
3	What is an interface? Explain it with example.	06	CO2	BTL4

Q.3	**Answer the following: (Any Two)**	**12**		
1	What are some common methods available in the Collection interface? Discuss.	06	CO2 CO3	BTL2
2	Explain implementation of ArrayList class in java and discuss how does it differs from Array.	06	CO2 CO3	BTL3
3	Explain URLConnection and HttpURLConnection.	06	CO4	BTL1
Q.4	**Answer the following: (Any One)**	**06**		
1	What is functional interface? How does Lambda expression is used to implement it? Explain.	06	CO2, CO4	BTL3
2	What is Lambda expression? What are the different types of it? Discuss.	06	CO2 CO4	BTL2

**** End of the Paper ****

Enrollment No._____

GANPAT UNIVERSITY

B.Sc. IT(CS) Sem-III Regular Nov-Dec 2024 (Remedial)

U63A4CN1: COMPUTER NETWORK-I

Time: 3 Hours

Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BTL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2-Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy**
1	Differentiate between wired and wireless computer networks.	06	CO1	BTL2
2	Compare ad-hoc and infrastructure modes in wireless networking.	06	CO2	BTL2
3	Illustrate the role of the subnet mask in IP addressing.	06	CO3	BTL3
4	Describe different network topologies.	06	CO1	BTL2
5	Explain multiple access protocols such as ALOHA and CSMA. How do they work in ensuring access control in a network?	06	CO2	BTL4
6	Compare the functions of UDP and TCP in network communication.	06	CO3	BTL3
7	Analyze the role of firewalls in network security and explain how they protect against cyber threats.	06	CO4	BTL1
Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy**
1	Explain the Main Use of computer networks.	06	CO1	BTL1
2	Evaluate TCP/IP reference Model.	06	CO1	BTL4
3	Write a short note on the Sliding Window Protocols.	06	CO2	BTL3
Q.3	Answer the following: (Any Two)	**12**		
1	Explain circuit switching and packet switching.	06	CO3	BTL5
2	Discuss the process of TCP connection establishment.	06	CO4	BTL2
3	Discuss the purposes of ARP and RARP protocols in network communication.	06	CO4	BTL1
Q.4	**Answer the following: (Any One)**	**06**		
1	Explain the techniques of cryptography and their importance in securing communication.	06	CO4	BTL6
2	Discuss the functions of SNMP, SMTP, and POP3 protocols.	06	CO4	BTL2

**** End of the Paper ****

Enrollment No._____

GANPAT UNIVERSITY

B.Sc.IT (CYBER SECURITY) Sem-III Reg\Rem Nov-Dec 2024

U63A5DF1: DIGITAL FORENSICS – I

Time: 3 Hours | Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BTL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2-Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy**
1	Analyze digital forensics' role in incident response.	06	CO1	BTL4
2	Justify the need for an incident response framework and identify key components.	06	CO1	BTL5
3	Explain how a network diagram supports evidence collection.	06	CO3	BTL3
4	Define evidence volatility and its role in evidence collection.	06	CO4	BTL2
5	Compare dead vs. live imaging and recommend suitable scenarios.	06	CO5	BTL4
6	Discuss the value of non-volatile data in investigations.	06	CO4	BTL5
7	Evaluate digital forensics' impact on cybercrime and legal support.	06	CO2	BTL6
Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy**
1	Describe key stages in incident response and explain their role in managing security incidents.	06	CO1	BTL2
2	Identify criteria for admissible digital evidence, and explain how forensic experts ensure compliance.	06	CO2	BTL3
3	Explain evidence preservation in digital forensics and describe techniques to prevent data alteration.	06	CO2	BTL3
Q.3	**Answer the following: (Any Two)**	**12**		
1	Describe the role of logs and log management in network forensics. What are the critical aspects to consider when managing logs for evidence collection?	06	CO3	BTL2
2	Discuss the types of network device evidence that can be collected during a network investigation. How do switches, routers, and firewalls contribute to evidence gathering?	06	CO3	BTL5
3	Explain the importance of preparation in host-based evidence collection. What are the key steps that must be taken before evidence acquisition begins?	06	CO4	BTL3
Q.4	**Answer the following: (Any One)**	**06**		
1	What is forensic imaging, and why is it crucial in digital forensics? Discuss how it preserves the integrity of digital evidence.	06	CO5	BTL5
2	Provide an overview of forensic documentation. Why is thorough documentation necessary for both technical and legal aspects of a forensic investigation?	06	CO5	BTL3

**** End of the Paper ****

Enrolment No._____

GANPAT UNIVERSITY

B.Sc.IT(CS) Sem-III Reg/Rem Nov-Dec 2024

U63B6EDM: ENVIRONMENT DISASTER MANAGEMENT

Time: 3 Hours

Instructions: Total: 60 Marks

1. Figures to the right indicate full marks, Course Outcomes and BTL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern:BTL1-Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy**
1	Explain the scope of environment.	06	CO1	BTL4
2	Define: Producer, Consumer and Decomposer.	06	CO2	BTL3
3	Explain the common sources of water pollution? How can we control water pollution?	06	CO3	BTL4
4	Define the following. (i) Hazard (ii) Disaster (iii) Vulnerability (iv) Risk (v) Coping Capacity (vi) disaster management cycle	06	CO3	BTL2
5	What do you mean by man-made (human aided) disasters?	06	CO5	BTL4
6	Explain the importance of planning in disaster management.	06	CO4	BTL3
7	Give the classification of ecosystem and explain in brief.	06	CO2	BTL4
Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy**
1	Explain any three positive and negative impact of technology development on environment.	06	CO1	BTL4
2	What are biotic and abiotic elements of environment? Explain with the examples.	06	CO2	BTL4
3	Explain the scope of environment.	06	CO1	BTL5
Q.3	**Answer the following: (Any Two)**	**12**		
1	Explain the common sources of air pollution? How can we control air pollution?	06	CO3	BTL3
2	Explain the common sources of noise pollution? How can we control noise pollution?	06	CO3	BTL3
3	Give the classification of Hazards. List and brief five hazards in each of the category.	06	CO4	BTL4
Q.4	**Answer the following: (Any One)**	**06**		
1	How do social, economic, and cultural factors influence disaster recovery efforts?	06	CO5	BTL4
2	How can insurance companies, police, and media collaborate effectively to enhance disaster preparedness and recovery efforts in local communities?	06	CO5	BTL4

**** End of the Paper ****